



Next Year in Futurism

Why the future keeps arriving late, unevenly, and in forms its prophets did not predict

The permanent near future

Some inventions seem to live forever five or ten years away. Fusion will soon provide abundant clean power. Cars will fly. Graphene will transform every product. Virtual reality will replace ordinary screens. Artificial general intelligence will perform any intellectual task. The date moves, but the confidence survives.

That does not make every delayed technology fraudulent. A sound scientific principle can be extraordinarily hard to turn into a reliable, affordable, regulated system. A prototype may work while manufacturing fails. A useful product may arrive without becoming universal. A forecast may identify the direction of change and still be badly wrong about speed, access, or social response.

Next Year in Futurism teaches forecast literacy. Its official links are evidence, not decorations around a list of funny predictions. They allow scholars to compare the language of a promise with what was actually built, who funded it, what obstacle appeared, and whether "arrival" meant a scientific milestone or everyday adoption. Facts about fast-moving fields such as AI, quantum computing, longevity, and cybersecurity also need dates. This guide reports what the official sources established and identifies claims that must be checked again before an event.

Chapter 1 - What futurism does

Futurism is not one method. It can be scientific modelling, business forecasting, political planning, speculative design, fiction, advertising, or wishful thinking. All these forms describe a world that does not yet exist, but they serve different audiences. A climate scenario asks what follows from assumptions. A startup pitch tries to attract money. A science-fiction story can test a moral possibility without claiming it will occur. Confusing those purposes produces bad evaluation.

Nine technologies that are "almost there"

Fusion energy seeks to release energy by joining light atomic nuclei. Experiments have achieved major scientific milestones, but a power system must produce useful electricity reliably, handle extreme materials conditions, breed or obtain fuel where required, maintain equipment, and cost enough to compete. "Net gain" can refer to energy at the target, the machine, or the whole facility. Each boundary gives a different headline.

A "cure for cancer" is difficult partly because cancer is not one disease. Tumours arise through different mutations, tissues, immune interactions, and stages. Some cancers are often curable; others remain much harder. Better prevention, screening, surgery, radiation, targeted therapy, and immunotherapy can transform outcomes without one universal cure arriving.

Graphene is a one-atom-thick carbon material with unusual strength, conductivity, and other properties. Producing small high-quality samples is different from manufacturing large, consistent, inexpensive sheets and integrating them into products that outperform existing materials. Nanotechnology has already arrived in particular coatings, electronics, medicine, and manufacturing, but the dramatic image of general-purpose molecular machines remains largely speculative.

Flying cars now appear more often as advanced air mobility: electric vertical-takeoff aircraft designed for limited routes. The engineering vehicle is only one piece. Noise, weather, certification, pilot or autonomy standards, air-traffic management, energy, vertiports, emergency landing, and price determine whether a demonstration becomes transport. Virtual reality likewise exists, but use is constrained by comfort, cost, isolation, content, and the fact that many tasks work better on an ordinary screen.

A space elevator would raise material along a tether extending beyond geostationary orbit. The concept promises repeated access without launching every kilogram by rocket, but Earth-based versions require an immensely strong, light, defect-resistant material and protection from weather, debris, vibration, and sabotage. Food pills can deliver nutrients or medicine; they have not replaced meals because food supplies water, bulk, pleasure, culture, and complex nutrition. Artificial general intelligence remains a disputed goal whose definition and test are less settled than confident timelines suggest.

Arrival has a ladder, not a switch

These nine examples become easier to compare if arrival is divided into stages. First comes a plausible principle. Then a repeatable experiment, a prototype outside the laboratory, a manufacturable product, regulatory approval, an affordable service, and ordinary use. A technology can climb one rung and remain years from the next. It can also skip the form imagined by its early promoters. Nanotechnology did not arrive as tiny universal robots, but nanoscale engineering already appears in chips, coatings, sunscreens, catalysts, and drug delivery. Virtual reality did not replace physical life, but it serves particular games, simulations, design tasks, and training environments.

Fusion shows why the rung must be named. Lawrence Livermore National Laboratory reported that the National Ignition Facility achieved ignition for an eleventh time on 20 June 2026, with a measured fusion yield of 7.9 megajoules and an approximate target gain of 3.8. That is a major and repeatable target-level result. It is not a power station: the lasers and facility require far more energy, and a commercial plant would need to fire, recover energy, replace targets, handle materials, and operate economically. Saying "fusion arrived" is defensible only if the speaker completes the sentence: arrived as what?

In a Scholar's Bowl round, that distinction protects against tempting distractors. If a question says a field has achieved a landmark experiment, the answer should not automatically be the option claiming mass adoption. If it describes an existing niche product, do not select the choice saying the original dream was fully realized. The curriculum is testing the distance between proof, product, system, and social habit.

Forecasts reveal their present

The official [L221 - 2018 forecast for 2028](#) is especially useful in 2026 because its target date is close enough to audit. It anticipated new mobile form factors, large numbers of autonomous miles, a more freelance workforce, disappearing cash, solar expansion, altered sports, security drones, vegetarian mainstreaming, VR addiction, and continued Brexit negotiation.

Do not mark each item simply true or false. Foldable phones, contactless payment, solar power, gig work, driver assistance, plant-based products, drones, and VR all exist. The forecast can still be wrong about scale, cause, geography, timing, or which version mattered. A prediction that "cash disappears" fails if digital payment grows while cash remains essential for resilience or inclusion. A prediction about autonomous miles needs to define supervision, road type, commercial fleet, and jurisdiction.

Adoption usually spreads unevenly rather than arriving everywhere on launch day. Early users tolerate price and inconvenience because the novelty has special value to them. Later users need reliability, compatible infrastructure, service, and a clear advantage over what they already own. Geography and income can put different populations on different parts of the curve. In 2028, one city might run a limited autonomous shuttle while most drivers elsewhere still control their own cars. Calling either "the status of autonomous transport" would hide the distribution.

The nine almost-there examples also belong to different forecast types. Fusion and a space elevator confront demanding physical and engineering limits. Graphene and nanotechnology often face manufacturing and integration problems. Flying cars and VR add regulation, infrastructure, comfort, and public acceptance. Cancer treatment and longevity confront biological diversity. Food pills conflict with the social functions of eating. AGI combines an unsettled definition with technical and political uncertainty. Their shared futurist glow should not erase those distinct bottlenecks.

Futurism organizes action before certainty

A forecast can change the future it describes. Governments prepare infrastructure, companies invest, students choose careers, and consumers delay or accelerate purchases. A frightening scenario can help prevent itself. A confident promise can attract the money needed to make itself more likely. It can also create a bubble that diverts resources from quieter solutions.

Forecasts can be self-defeating too. A warning that encryption will become vulnerable may cause migration to safer standards, so the predicted disaster never occurs. That does not prove the warning was wrong. Conversely, a forecast that everyone will use a platform can encourage developers and businesses to support it, helping create the network effect it predicted. Evaluation must ask whether the prediction merely described behaviour or helped cause the outcome.

This is why futurism deserves study even when it is wrong. It identifies what a society wants, fears, and believes can be engineered. The 2018 article treats mobile devices, autonomy, flexible work, payment, diet, and entertainment as places where technology and markets will reorganize daily life. It pays less attention to political rupture, public health, war, or the possibility that people may resist a technically available system.

For *Are We There Yet?*, futurism supplies the destination before the road has been built. A good scholar asks who drew it, what evidence supports the travel time, and whether arriving there would solve the original problem.

Sources for this chapter: The direct official source is [L221 - "Take me to the Year 2028"](#). Its value is the dated forecast, not authority over every field it mentions. The current fusion example comes from the [National Ignition Facility's experiment record](#), reviewed in July 2026. Use primary scientific or government sources for cancer, advanced air mobility, and materials, and state which meaning of "arrival" is being measured.

Chapter 2 - Hype cycles

A hype cycle begins when a result attracts expectations faster than a working system can grow. Early successes are real, but media, investors, and institutions extrapolate them. Problems then look like betrayal. Attention collapses, even while slower technical work continues. Some technologies later reach useful forms. Others remain narrow or disappear.

Wardenclyffe: a prototype for a system that did not exist

In 1904, Nikola Tesla said it would soon be possible for an individual to transmit wireless messages around the world with simple apparatus. The sentence sounds like a smartphone prediction. It was connected to Tesla's own project and should not be treated as a detached prophecy. The [L222 - Tesla Science Center history of Wardenclyffe Tower](#) describes the Long Island facility built from 1901 onward for wireless communication and Tesla's more ambitious ideas about transmitting power.

Tesla's vision contained genuine insight. Wireless global communication did become ordinary. The route was not the one tower and funding model he imagined. Radio engineering, spectrum regulation, networks of transmitters and receivers, electronic components, satellites, cables, computing, standards, manufacturing, and decades of investment built the eventual system. Similarity at the level of desire does not prove that Tesla had designed the smartphone.

Wardenclyffe struggled for funding as priorities and competing systems changed. J. P. Morgan's support was not infinite, costs rose, Marconi demonstrated wireless telegraphy through another approach, and the tower never became the promised global service. The lesson is not that institutions crushed a completed miracle. A large system needs technical proof, capital, a service people will pay for, governance, and compatibility. Vision can be ahead of one of those and behind another.

Tesla also blurred communication and power transmission in ways that made the business case harder to evaluate. A financier could support a service that charges customers for messages. A worldwide system whose power or information could be received broadly raised harder questions about control and revenue. The tower thus failed at the intersection of experiment and institution. It was not enough for parts of the science to be promising; the proposed network needed an operator and a path from demonstration to repeatable service.

"Soon" hides several clocks

The curriculum invites scholars to search sentences beginning "it will soon be possible" or "in a few years, everyone will." The three starter claims move from a 1925 promise that one person could be heard by every human, through a 2019 proposal for a satellite that makes its own solar array in orbit, to a 2025 longevity claim that humans might live for a thousand years.

Each needs separate clocks. Scientific possibility asks whether known physics permits the outcome. Engineering asks whether a prototype works under controlled conditions. Manufacturing asks whether it can be produced consistently. Economic readiness asks whether benefits exceed costs for an actual customer. Regulatory readiness asks whether the risk is accepted. Adoption asks whether people change habits and whether infrastructure reaches them. "Soon" can be accurate on one clock and absurd on another.

Quantum supremacy is a milestone, not a product launch

Google's 2019 experiment used a quantum processor to perform a specialized sampling task. The [L226 - quantum supremacy report](#) describes the headline and IBM's challenge to the classical-computing comparison. The phrase means that a quantum device performed a particular computation beyond the practical reach claimed for a classical computer. It does not mean the machine was better at word processing, logistics, weather, or every calculation.

Quantum bits can exploit superposition and entanglement, but they are fragile. Noise and error correction create enormous engineering demands. The [L227 - June 2025 applications report](#) described limited practical use and strong investment, especially around chemistry and physics simulation. That dated assessment should not be converted into

the timeless statement "quantum computers have no purpose." Research milestones can change quickly, and usefulness differs between laboratories and ordinary consumers.

Preparing for Q-Day before it arrives

Q-Day is the feared point when a cryptographically relevant quantum computer can break widely used public-key systems. The [L228 - Q-Day investigation](#) dramatizes the risk. The important policy response begins before a date can be predicted. Sensitive data intercepted now might be stored and decrypted later. Organizations need inventories of cryptographic systems, migration plans, new standards, and enough time to update hardware and software.

This reverses the usual hype problem. Waiting until the technology visibly arrives may be irresponsible because migration takes years. At the same time, fear can sell products and inflate timelines. The disciplined response is neither panic nor dismissal. Identify the capability required, estimate exposure, prepare reversible steps, and update as evidence changes.

Who benefits from the peak of expectation?

A founder can raise capital, a research institution can attract grants, a journalist can earn attention, and a government can display national ambition. None of those incentives proves a claim false. They explain why uncertainty may be compressed into a confident date. The critic has incentives too: a dramatic debunking earns attention and can mistake an early limitation for a permanent one.

A forecast audit should therefore preserve the original wording and date, identify the speaker's role, and record what would count as success. It should revisit the claim at fixed intervals rather than only after a famous failure. Moving goalposts become visible when "a practical quantum application" turns into "a useful research experiment" after the deadline, or when a prototype is retrospectively treated as mass adoption.

Readiness can arrive before the threat

Post-quantum cryptography is a useful counterexample to the idea that preparation proves a prediction. In August 2024, the US National Institute of Standards and Technology finalized its first three post-quantum cryptography standards and encouraged administrators to begin migrating. That decision did not certify that a code-breaking quantum computer was ten years, five years, or one year away. It recognized that cryptographic transitions are slow, old systems are hard to inventory, and some secrets must remain protected for decades.

The action is valuable under several futures. If Q-Day comes earlier than expected, migration reduces exposure. If it comes much later, replacing brittle cryptography and building the ability to change algorithms can still improve security. This is a robust forecast response: it creates benefits across a range of dates rather than betting everything on one countdown. By contrast, buying a mysterious product advertised as "quantum-proof" without knowing its algorithm, implementation, or update path converts fear into a market.

A round may pair two statements that sound contradictory: quantum computers have few broad commercial applications, and governments are already changing encryption. Both can be true because the clocks differ. The first concerns useful computation now; the second concerns the time needed to protect long-lived data. Strong teams identify the decision horizon before judging whether preparation is premature.

Hype cycles occur because the public asks a binary question: is the future here? Research moves through partial, field-specific milestones. The better question is what became possible, under which conditions, for whom, and what remains between the demonstration and a durable system.

Sources for this chapter: Wardencllyffe is anchored by [L222](#). The official quantum trail is [L226](#), [L227](#), and [L228](#). The migration case uses [NIST's August 2024 standards announcement](#). Date all readiness claims and distinguish a benchmark, useful application, broad adoption, and prudent preparation.

Chapter 3 - False imminence

False imminence is the feeling that a dramatic outcome is close because its components have become vivid. A laboratory can grow an organoid. A surgeon can transplant an organ. A company can reprogram cells. A public figure can speak about living to 150. These facts make radical life extension feel like one final assembly step, even when the missing steps include the central problem.

Extending life is not one technology

The official [L223 - report on Xi Jinping and Vladimir Putin discussing longevity](#) begins with a 2025 hot-mic conversation about repeated organ replacement. Organ transplantation already saves and extends lives. It also requires a suitable organ, difficult surgery, management of rejection, and long-term care. Replacing organs one by one does not stop aging across the brain, immune system, blood vessels, and other tissues.

Organoids are small three-dimensional tissue models grown from cells. They can help researchers study development, disease, and treatment. They are not miniature complete replacement organs. Bioprinting places cells and biomaterials in designed structures, but making thick, vascularized, innervated tissue that functions safely for years is far harder than printing a shape.

Plasma interventions arose partly from experiments in which young and old animals shared circulation. Translating those signals into safe human rejuvenation has proved difficult, and commercial claims can move ahead of evidence. Senotherapeutics aim to target cellular senescence, a state in which damaged cells stop dividing and may release harmful signals. Clearing such cells or altering their effects might improve particular age-related conditions, but aging is not caused by one cell type alone.

Cellular reprogramming can reset aspects of a cell's identity and biological age. Partial reprogramming tries to capture rejuvenating effects without erasing cell identity or creating tumours. Cellular rejuvenation is a broad goal rather than one established treatment. Telomeres protect chromosome ends and shorten in many dividing cells, but extending them indiscriminately can affect cancer risk. Caloric restriction extends lifespan in several organisms and affects metabolism, yet human benefit, adherence, nutrition, and quality of life complicate the translation.

Cryonics preserves a legally dead body or brain at low temperature in the hope that future technology can repair damage and restore the person. No cryonically preserved human has been revived. Digital immortality might preserve recordings, writing, preferences, or a model that imitates someone. Whether such a model is the same person is a philosophical and legal question, not an engineering detail.

An evidence ladder for longevity claims

The words "studied," "tested," and "available" are easy to mistake for "shown to extend human life." A mechanism in cells is weaker evidence than a result in animals. An uncontrolled treatment report is weaker than a randomized human trial. A change in a biomarker is not automatically a reduction in illness or death. Trial registration proves that a study was listed, not that the treatment works. Regulatory approval for one medical purpose does not approve a product for rejuvenation.

This ladder changes how two curriculum examples should be read. In December 2024, the US Food and Drug Administration repeated that young-donor plasma was not approved to prevent normal aging or memory loss and that it knew of no demonstrated clinical benefit for those uses. The National Institute on Aging has reported a much narrower human finding for caloric restriction: a two-year trial found a small change in one measure of the pace of biological aging, while longer follow-up and other measures were still needed. Neither result supports "immortality is close." One warns against a commercial leap; the other shows how cautious progress is actually described.

Longer life for whom?

A longevity technology can work medically and fail socially. If access follows wealth or political power, longer lives could let elites retain property and office for far longer. Pension, inheritance, housing, education, family structure, and

intergenerational authority would change. Extra years in poor health are not the same outcome as longer healthspan, the period lived without severe disease or disability.

Researchers can measure lifespan, disease-free survival, functional ability, biological markers, and reported quality of life. These outcomes are related but not identical. A treatment that changes a biomarker without helping people function has not established rejuvenation. An intervention that delays one disease may expose another later. Longevity headlines often move from an animal marker to a human-life claim without showing the intermediate evidence.

The ethics cannot be settled by claiming that death gives life meaning. Preventing early death and reducing suffering are central achievements of medicine. Nor can it be settled by saying more life is always better. Treatment burden, consent, environmental demand, unequal access, and the distribution of care all matter. The first policy goal may be extending healthy life for many rather than maximum lifespan for a few.

AI detection offers another false finish line

The curriculum then turns from biological identity to authorship. Wikipedia's [L229 - field guide to signs of AI writing](#) documents patterns editors have noticed in generated contributions: generic importance claims, repetitive summaries, formulaic contrast, invented references, awkward headings, and other stylistic or formatting clues. It is a practical list for a particular editing environment, not a scientific detector.

Human writers can use the same habits. Models and users can change prompts to avoid them. A detector that assigns a probability can produce false accusations, especially against writers using predictable school forms or writing in an additional language. Style can support suspicion; evidence about process, sources, drafts, and understanding must carry the judgment.

The [L230 - professor's account of setting an AI trap](#) describes deliberate bait placed where generated answers might retrieve it. Such a method can reveal dependence in a particular assignment, but it does not solve assessment. Traps can become adversarial, punish careless research alongside AI use, and encourage a contest over concealment.

Authorship policy also has to distinguish prohibited substitution from permitted assistance. Spellcheck, translation, brainstorming, feedback, retrieval, and full answer generation do not place the same cognitive work outside the student. A course should say what process it wants to assess and what disclosure is required. Otherwise students receive a rule about a tool when the real concern is evidence of thinking.

A better question is what the assignment should demonstrate. If students must evaluate a source, explain a decision, answer questions orally, preserve notes, and revise from feedback, a polished final paragraph cannot substitute for the whole process. The educational destination is understanding, not text that merely looks human.

The same evidence ladder applies here. A highlighted phrase is a clue, not proof. A detector score is a measurement produced under particular assumptions, not a verdict. A fabricated citation, when checked against the student's notes and explanation, is stronger evidence of an unreliable process. A student who can defend choices, locate sources, and revise accurately may demonstrate learning even if a tool assisted part of the work. Assessment should test the claimed human contribution rather than pretend authorship can always be inferred from punctuation.

Longevity and AI authorship share a structure. Visible components create the impression that the final problem has nearly disappeared. An organ model is not indefinite human renewal. Fluent prose is not verified understanding. False imminence collapses the distance between resemblance and function.

Sources for this chapter: The longevity branch begins with [L223](#). The evidence examples use the [FDA's December 2024 young-plasma warning](#) and the [National Institute on Aging's account of the CALERIE analysis](#). Every listed method needs a current biomedical source because evidence and trial status change. The AI-authorship branch is [L229](#) and [L230](#). Neither link establishes a reliable universal detector.

Chapter 4 - Retrofuturism as evidence

Retrofuturism looks at futures imagined in the past. The easiest reaction is laughter: people drew personal helicopters, moon Olympics, moving sidewalks, domestic robots, and streamlined cities that never appeared as advertised. The harder and more useful reaction is historical. Why did that future look desirable, and what part of the present was projected forward?

Old futures preserve old priorities

The official gallery of [L224 - retro-future predictions](#) collects images from the twentieth century, while the [L225 - decade-by-decade prediction video](#) shows how expectations changed. The images often preserve the social structure of their own time inside new machinery. A kitchen becomes automated but a woman remains assigned to it. Transport becomes faster while cities remain organized around older assumptions. Space inherits the competition and spectacle of Earth.

Some predictions are directionally close. Video calls, automatic doors, remote shopping, smart-home controls, high-speed transport, and portable media all have ancestors in older visions. Their real forms arrived through networks and ordinary devices rather than one dramatic machine. A smartphone absorbed camera, map, ticket, newspaper, music player, television, wallet, and communicator, which is harder to depict than a single futuristic car.

To read an old future as evidence, first date and locate it. Identify who commissioned the image, who was expected to see it, and whether it sold a product, entertained readers, or argued for policy. Then list what changed and what remained familiar. Clothing, household roles, street layout, race, class, energy, and labour often reveal more about the forecast's present than the featured machine does.

The method avoids hindsight arrogance. A prediction might be reasonable given information available at the time even if an unexpected war, regulation, discovery, or market defeats it. Another might become true only because the predicted deadline mobilized work. Accuracy matters, but historical quality also depends on assumptions, uncertainty, and the decision the forecast supported.

Other forecasts failed because they optimized a visible inconvenience while ignoring behaviour and systems. Food pills remove cooking and chewing but also remove pleasure, tradition, texture, and social meals. Nuclear cars promised extraordinary range while hiding shielding, crash risk, waste, and reactor control. A moon Olympics imagines that transport and life support become easy enough for spectacle, then imports an existing institution instead of asking what low gravity and isolation would make culturally new.

A forecast can be wrong and still cause change

Visions recruit engineers, investors, voters, and children. A drawing of an automated home can shape product goals even when its exact machines fail. Science fiction gives names and images to possibilities before policy has language for them. Fearful futures can produce safety research or restrictive law.

Retrofuturism also reveals forgotten alternatives. Once a society builds around private cars, it can treat that route as inevitable and forget futures based on public transit, compact cities, or different working patterns. An archive reminds us that roads not taken were imaginable. Today's future is likewise one selection among many, not the automatic result of technology.

Read a future image at three levels

Begin with the object: what machine, building, or practice is shown? Then inspect the system it assumes. A personal aircraft requires landing space, fuel or charging, traffic rules, maintenance, weather decisions, and a way to prevent crashes above homes. Finally inspect the social script. Who works, who is served, who owns the device, and who remains outside the frame? This method turns a charming illustration into evidence about institutions.

It also prevents smug hindsight. A prediction can miss the object and identify the need. The household videophone was often drawn as a special appliance, but face-to-face calling arrived inside multipurpose computers and phones. A

forecast can get the object right and miss the social result. Remote work technology existed before it became normal for many workers, because organizational permission and a crisis mattered alongside bandwidth. Accuracy has several dimensions: function, form, date, scale, and consequence.

The frontier can point downward

The Kola Superdeep Borehole is not a fantasy image. It is a real project that acquired a futuristic aura through scale and Cold War rivalry. The United States began Project Mohole with the ambition of drilling through the ocean crust toward the mantle, but the effort was cancelled after technical, management, and funding difficulties. The Soviet Kola project began drilling in 1970 and eventually reached 12,262 metres, the deepest artificial vertical borehole.

The [L232 - history of the deepest drilling race](#) explains how heat became a major barrier. Rock at depth was hotter and behaved differently from predictions, making further drilling difficult. The borehole remained only a tiny fraction of the distance to Earth's centre. A kilometre sounds impressive until the chosen destination is planetary.

The [L233 - Orion essay on Kola](#) adds place, environmental damage, Indigenous Sami land, industrial history, and the remains of the site. This perspective matters. A frontier story can describe empty territory waiting for human achievement while ignoring people and ecosystems already there.

The drilling also generated scientific knowledge despite missing the imagined destination. Samples and measurements changed understanding of crustal composition, water at depth, temperature, and the behaviour of rock under extreme conditions. "They did not reach the mantle" is true but incomplete. A project can fail according to its headline goal and still produce valuable intermediate findings. The value should be documented rather than assumed as consolation.

Kola is a useful transfer to space missions and giant scientific instruments. Ambitious goals can justify tools whose discoveries cannot be predicted in detail. That argument needs limits: open-ended curiosity does not remove duties to budget, affected communities, environmental repair, or less glamorous science. The frontier is a choice among uses of scarce attention.

Rivalry accelerates and distorts

Competition can concentrate money, talent, prestige, and urgency. The space race produced rapid achievements. It also duplicated effort, tolerated risk, and directed science toward national display. Cooperation can share cost and data but move slowly through coordination and political disagreement.

George Mallory's phrase "because it's there" gives exploration a pure motive. The mountain or depth appears to issue a challenge. In practice, frontiers are chosen. A society can spend on drilling, space, ocean research, disease prevention, clean water, or basic maintenance. Curiosity is a real good, but "there" does not decide its own budget.

For a round, compare the retrofuture gallery and Kola before choosing an answer. The gallery shows imagined destinations whose infrastructure was missing; Kola shows a funded machine that met unanticipated geology and heat. Both expose assumptions, but only one is a completed historical experiment with measurable depth. A distractor may borrow the lesson of one case and apply it too broadly to the other. Ask whether the evidence is an image, a promise, a prototype, or an operating project.

Retrofuturism teaches humility without cynicism. Old forecasts look naive partly because we know which branch history took. Our own images contain the same blind spots. The right response is to preserve multiple futures, state assumptions, and ask what social order each machine quietly carries inside it.

Sources for this chapter: The official retrospective sources are [L224](#) and [L225](#). The underground frontier is [L232](#) and [L233](#). Use Kola's verified depth rather than repeating myths about sounds from hell.

Chapter 5 - AI, prediction bias, and hope under pressure

Artificial general intelligence is usually defined as a system able to learn, understand, and apply knowledge across a very broad range of tasks at human-like or greater flexibility. Today's systems can generate language and images, write code, recognize patterns, and perform strongly on many evaluations. They can also hallucinate facts, fail on unusual cases, inherit biases from data and feedback, and produce a convincing answer without a stable model of the situation.

Performance is broader than one benchmark

A narrow system can exceed human performance in one domain without being generally intelligent. A calculator's arithmetic does not imply judgment. A chess engine's strength does not let it cook. Modern foundation models are broader, since one model can perform many language and media tasks. Breadth alone does not settle whether the system understands, can learn autonomously in the world, or should count as AGI.

Benchmarks also change behaviour. Once a test becomes important, developers train toward it, examples may leak into data, and the score becomes less independent. Real environments contain outliers, shifting goals, missing information, adversaries, and consequences that a test set may not represent. Reliability in the long tail can matter more than average fluency.

General intelligence also includes transfer and repair. A system that succeeds only when the prompt resembles training examples may be broad in output but brittle in adaptation. A capable agent should notice that a plan failed, seek missing information, revise without losing the goal, and understand when it should defer. Those abilities are difficult to measure because a benchmark usually fixes the task before the system begins.

The 2026 evidence is strong and jagged

The [International AI Safety Report 2026](#) gives a more current baseline than a single product launch. Its expert group reported continued gains in mathematics, coding, science, and some autonomous operation, driven partly by post-training methods and more computation while producing an answer. It also described capability as "jagged": leading systems can solve difficult benchmark problems and then fail at a simpler physical, visual, or multi-step task.

That pattern matters for the word "general." A long list of impressive skills does not prove stable competence across unlimited contexts. Nor does one embarrassing failure prove that the system has no broad value. The right test samples unfamiliar tasks, measures recovery from error, records how much human supervision was needed, and follows performance after deployment. July 2026 evidence supports rapidly improving general-purpose systems. It does not supply a universally accepted AGI finish line.

Scaling is evidence, not destiny

The official [L231 - Science Focus article on the AI bubble and AGI limits](#) challenges the assumption that larger models, more data, and more computing must produce general intelligence. Scaling has yielded substantial gains. It also raises energy, chip, data, capital, and evaluation costs. Returns may continue, slow, or require new architectures and training methods.

Economic expectation can move faster than technical value. Companies invest because competitors invest. Infrastructure is built for demand forecasts that may not materialize. At the same time, calling the field a bubble does not imply the technology disappears. Railways, telecommunications, and the internet all experienced speculation alongside durable change. Financial timing and technical usefulness are different questions.

Should we want to arrive at AGI?

Supporters imagine faster science, personalized education, improved medicine, abundant expertise, and relief from dangerous or dull work. Critics fear labour displacement, concentrated power, surveillance, manipulation, autonomous weapons, loss of human control, and systems pursuing poorly specified goals. Both lists can become vague. A strong argument names a capability, institution, owner, affected group, and safeguard.

If a system can perform valuable cognitive work, who owns it and who receives the gain? If it advises a doctor or judge, who is accountable for error? If it creates persuasive media at scale, how is consent protected? If only a few states or firms can train the leading systems, technical arrival may increase political distance between people.

Work will not change through one replacement rate. Tasks within an occupation differ. A system may draft routine text while increasing demand for verification, client judgment, data preparation, and responsibility. Employers can use the same tool to reduce drudgery, intensify monitoring, deskill workers, or expand service. The institutional choice determines whether productivity becomes time, pay, access, or simply more output expected from fewer people.

Education faces the same fork. Cheap explanation and feedback could help a learner who lacks a tutor. Fluent answer production can also let a student pass through an assignment without the productive difficulty that builds knowledge. The useful design asks the model to support attempts, expose sources, and invite revision while preserving moments when the learner must retrieve, reason, and explain independently.

Hope is not foolish. It motivates research and makes alternatives imaginable. Under pressure, however, hope can become a deadline used to excuse weak evidence: competitors are racing, so safety must wait; the benefit will be enormous, so current harm is temporary; abundance is coming, so distribution can be solved later. Prediction literacy keeps the goal visible while refusing to spend people as fuel for it.

Scenario thinking is more honest than one countdown. In one future, scaling continues and general systems arrive quickly; governance must prepare for concentration and severe misuse. In another, progress slows but task-specific systems spread through institutions; mundane procurement, labour, privacy, and error become the main issues. In a third, public trust or resource constraints limit deployment. Decisions that are useful across several scenarios, such as evaluation, incident reporting, worker voice, and clear accountability, are stronger than bets that require one exact timeline.

Govern the route before agreeing on the destination

Society does not need a final definition of AGI before governing systems that already act. A hiring model can be audited for discrimination. A medical system can require clinical validation and a responsible professional. A model with access to email, money, or laboratory tools can receive tighter permissions, monitoring, and shutdown procedures. These controls follow actual capability and exposure rather than a dramatic label.

The 2026 report calls this an evidence dilemma. Acting on weak evidence can lock in ineffective rules, while waiting for conclusive proof can leave serious harms unmanaged. A practical answer uses staged commitments: test before deployment, increase safeguards as capability or access grows, record incidents, allow independent evaluation, and revise thresholds. This is how uncertainty becomes a design condition instead of an excuse for either paralysis or reckless speed.

In debate, avoid arguing "AGI will save us" against "AGI will destroy us." Those claims hide all the examinable parts. Name the system, task, failure, operator, affected group, and remedy. Then separate a current harm, a plausible near-term risk, and a speculative extreme outcome. The conclusion may still be cautious or ambitious, but it will rest on evidence rather than atmosphere.

A practical forecast audit

When a future claim appears, identify its exact outcome and date. Ask what stage the evidence supports: concept, laboratory result, prototype, regulated product, affordable service, or widespread use. Name the missing dependencies, including energy, materials, law, trust, training, maintenance, and demand. Look for a base rate: how long have similar systems taken? Search for incentives and disconfirming evidence. Decide what would make the forecast wrong. Return to it later.

Apply that method to every branch. Fusion is not "here" because one target produced a net-energy result. Quantum supremacy is not a general computer. An organoid is not an organ replacement. Human-like prose is not verified human authorship. A deep borehole is not access to the mantle. A prediction becomes useful when its destination and progress measures can survive that precision.

Sources for this chapter: The official AGI source is [L231](#), read beside the writing evidence in [L229](#) and [L230](#). The dated capability and governance baseline is the [International AI Safety Report 2026](#), which synthesizes evidence published before December 2025. AI capability claims can change within months. Treat July 2026 as the document's review date, not a permanent frontier.

Final synthesis - A future needs more than a countdown

The section does not teach that predictions are silly. It teaches that a prediction contains hidden definitions. Does arrival mean a physical effect, a prototype, a profitable product, safe regulation, equal access, or a change in ordinary life? Tesla was directionally brilliant about global wireless communication while wrong about the immediate system that would provide it. Quantum computing can reach a major benchmark while remaining impractical for most work. AI can write fluently while authorship and reliability remain unresolved.

Old futures help because their assumptions are easy to see after the fact. They projected familiar families, work, politics, and desires into new machines. Our forecasts do the same. Longevity can preserve unequal power. AGI can automate an institution's current values. A frontier can be called empty while people already live beside it.

The best answer to *Are We There Yet?* names the particular "there," the evidence available on a date, the dependencies still missing, and the people affected by the route. Hope can point toward a destination. It cannot substitute for a map.